

CSE4078 Spring 2026 Term Project

Progress Summary

Group 2

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1 Project Objective

The objective of this project is to evaluate the Turkish performance of small open-source Large Language Models (LLMs), compare their baseline performance, and select one model for further fine-tuning. The selected model will later be fine-tuned using the mandatory supervised fine-tuning (SFT) corpus assigned to our group.

2 Selected Baseline Models

The project requires each group to evaluate two small open-source LLMs in the approximately 0.3B to 4B parameter range before selecting one model for fine-tuning. It also recommends parameter-efficient fine-tuning methods such as LoRA or QLoRA and requires even-numbered groups to use the `Renicames/turkish-law-chatbot` dataset as the mandatory SFT corpus [1].

For the baseline evaluation stage, we selected **Qwen/Qwen2.5-1.5B-Instruct** and **meta-llama/Llama-3.2-1B-Instruct**. Qwen2.5-1.5B-Instruct was selected because it is an instruction-tuned model from the Qwen2.5 family, which includes models ranging from 0.5B to 72B parameters [2]. The Qwen2.5 technical report also describes improvements in pretraining and post-training for this model family [3].

Llama-3.2-1B-Instruct was selected as the second baseline because Meta describes the Llama 3.2 text models as pretrained and instruction-tuned multilingual models released in 1B and 3B sizes [4, 5]. These two models provide a fair comparison because both are small instruction-tuned models, but they come from different model families.

3 Datasets

3.1 SFT Corpus

Since our group number is even, the mandatory SFT corpus assigned to our group is:

This dataset contains Turkish legal question-answer pairs. The Hugging Face dataset card identifies it as a Turkish legal question-answering dataset with tags such as law, Turkish, constitution, and legal questions [6]. It will be used later for fine-tuning the selected model.

3.2 Evaluation Data

For the current progress stage, we evaluated the selected models on Turkish legal question-answering samples. The same 40 questions were used for both models in order to make the comparison fair.

4 Baseline Evaluation Setup

The models were evaluated using zero-shot inference. Each model received the same Turkish legal questions with a system prompt instructing it to act as a Turkish law assistant. The generated answers were compared with reference answers.

We used ROUGE-1, ROUGE-2, and ROUGE-L to measure lexical overlap between generated and reference answers [7]. In addition, Exact Match and Token F1 were used as answer-level and word-level comparison metrics. Since legal answers may be correct even when phrased differently, the automatic metrics were supported with manual inspection for hallucinations, incomplete answers, and legally inaccurate statements.

The models were loaded using the Hugging Face Transformers library [10], and the dataset was accessed using the Hugging Face Datasets library [11].

5 Baseline Results

Table 1 shows the baseline results on the same 40 question-answer samples for both models.

Table 1: Main baseline evaluation results

Model	ROUGE-1	ROUGE-2	ROUGE-L	Token F1	Exact Match
Qwen2.5-1.5B-Instruct	0.2471	0.0754	0.1574	0.1651	0.0000
Llama-3.2-1B-Instruct	0.1696	0.0597	0.1287	0.1361	0.0000

Qwen2.5-1.5B-Instruct achieved higher ROUGE and Token F1 scores than Llama-3.2-1B-Instruct. However, both models received an Exact Match score of 0.0000, which is expected in open-ended legal question answering because generated answers often differ from the reference wording.

Table 2: Additional baseline output statistics

Model	Avg. Pred. Words	Avg. Ref. Words	Suspicious Ratio	Suspicious Count
Qwen2.5-1.5B-Instruct	135.75	54.63	0.60	24
Llama-3.2-1B-Instruct	99.35	54.63	0.00	0

Qwen generated longer answers on average, which sometimes improved overlap with reference answers but also increased the risk of unsupported explanations. The suspicious-answer flag is only a simple rule-based indicator, so manual inspection is still necessary for evaluating legal correctness.

6 Example Error Analysis

Manual inspection showed that both models could generate fluent Turkish text, but they did not always provide legally accurate answers. The most common problems were hallucination, overly general answers, incomplete answers, and incorrect legal statements.

For example, when asked about Article 1 of the Turkish Constitution, the expected answer should directly state that the form of the Turkish state is a republic. However, the models sometimes produced unrelated or invented explanations instead of the direct constitutional content.

Qwen2.5-1.5B-Instruct often generated longer and more detailed answers, but some of these answers contained unsupported legal claims. Llama-3.2-1B-Instruct generally produced shorter answers, but they were sometimes incomplete or unstable. These observations support the need for supervised fine-tuning on the assigned Turkish legal question-answering corpus.

7 Selected Model for Fine-Tuning

Based on the baseline results, we selected:

Qwen/Qwen2.5-1.5B-Instruct

Qwen was selected because it achieved better ROUGE and Token F1 scores than Llama and produced more usable Turkish answers. It is also small enough for parameter-efficient fine-tuning. We plan to use QLoRA, which combines quantized model weights with LoRA-style adaptation and is suitable for limited GPU memory environments [8, 9].

8 Fine-Tuning Plan

For the next stage, we plan to fine-tune Qwen/Qwen2.5-1.5B-Instruct using the `Renicames/turkish-law` training split. The test split will not be used during training and will be kept only for evaluation.

The planned workflow is:

1. Preprocess the SFT corpus.
2. Split the training data into training and validation subsets.
3. Fine-tune Qwen using QLoRA.
4. Re-evaluate the fine-tuned model on the same evaluation set.
5. Compare baseline and fine-tuned results.
6. Analyze remaining error patterns.

9 Conclusion

In this progress stage, we selected two small instruction-tuned LLMs and evaluated them on Turkish legal question-answering samples. Qwen/Qwen2.5-1.5B-Instruct performed better than Llama-3.2-1B-Instruct according to ROUGE-1, ROUGE-2, ROUGE-L, and Token F1 scores. However, both models showed legal accuracy issues, which supports the need for domain-specific fine-tuning. Therefore, Qwen/Qwen2.5-1.5B-Instruct was selected for the next stage.

References

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